

# Frontend Development and Flask Web Application Integration

## Overview

Three HTML pages are created and stored inside the **templates/** folder to build the frontend of the Credit Card Approval Prediction application. The **home.html** page acts as the landing page and provides a project overview along with a navigation button to access the prediction form. The **index.html** page contains the input form where users enter applicant financial and demographic details required for prediction. The **result.html** page displays the final prediction result showing whether the credit card application is approved or rejected.

In this milestone, a Flask web application is developed and integrated with the trained machine learning model to provide real-time prediction through a web interface. The Flask application acts as a bridge between the frontend HTML pages and the backend machine learning model. User inputs entered through the form are collected in the backend, processed, and passed to the trained model for prediction.

The **app.py** file is used as the main backend script of the Flask application. It handles routing, model loading, input processing, prediction generation, and result rendering. The trained model is loaded using Pickle or Joblib, and Flask routes are created to display the home page, prediction form, and result page. When the user submits the form, the entered values are processed and passed to the model using the **predict()** method to generate the final approval prediction.

## Workflow

1. User opens the Home page.
2. User navigates to the prediction form.
3. Applicant details are entered.
4. Flask receives and processes the data.
5. The trained model predicts approval status.
6. The result page displays Approved or Rejected.

## Outcome

The Flask application successfully integrates the frontend with the machine learning model, enabling users to obtain instant credit card approval predictions through a web-based interface.

## Figure: HTML Input Form (index.html)

```

2    <html lang="en">
242  <body>
244    <div class="container">
248      <section class="hero">
270        <div class="hero-card">
280          <input type="text" placeholder="Enter applicant name">
281        </div>
282
283        <div class="input-box">
284          <label>Income Type</label>
285          <select>
286            <option>Working</option>
287            <option>Commercial Associate</option>
288            <option>Pensioner</option>
289          </select>
290        </div>
291
292        <div class="input-box">
293          <label>Education Type</label>
294          <select>
295            <option>Graduate</option>
296            <option>Post Graduate</option>
297            <option>High School</option>
298          </select>
299        </div>
300
301        <div class="input-box">
302          <label>Annual Income</label>
303          <input type="number" placeholder="Enter income">

```